Example:

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5]
y = [10, 20, 25, 30, 35]
plt.figure(figsize=(8, 5)) # Adjusting the plot size plt.plot(x, y, marker='o', linestyle='--', color='blue')
plt.title("Data Trends", fontsize=14) plt.xlabel("X-axis", fontsize=12)
plt.ylabel("Y-axis", fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7) plt.show()
```

Features:

- Wide variety of plots, including line, bar, scatter, and histogram.
- Customizable visualizations with titles, labels, and annotations.
- Integration with Pandas and NumPy for efficient visualization workflows.

By mastering NumPy, Pandas, and Matplotlib, you can build a strong foundation in data analysis, which is essential for any ML or data science project. These libraries allow you to process data efficiently, uncover patterns, and present results effectively.

Writing Your First Python Script for Data Analysis

Writing Python scripts to analyze datasets is a foundational step in the Machine Learning (ML) journey. Data analysis serves as the gateway to understanding, processing, and deriving insights from raw data. Python, with its robust libraries, makes this task both efficient and approachable. Below is an expanded example demonstrating how to load, process, and visualize data using Pandas and Matplotlib.

Script to Analyze a Dataset

Step 1: Import Libraries

The first step in any Python-based data analysis script is importing the required libraries. Pandas is used for handling and processing data, while Matplotlib is employed for creating visualizations.

```
import pandas as pd
import matplotlib.pyplot as plt
```